Supplement, appendix and contribution statement

Everything here supports the main document; none of it changes a registered result. Post-hoc and synthetic analyses are labelled as such. Model names: B1c = per-state quantile rule with the calibration shift; C* = linear quantile regression; D* = shallow HGB; B0 = the project's retained rule (no shift).

S1. Model selection on validation data (F0 → V)

Metric: pinball loss at q = 0.90 on the unshifted integer promise $\max\{1, \lceil \hat{y} \rceil\}$. A linear step was adopted only if it lowered the loss by at least 1%. Candidates within 1.001 × the global minimum counted as tied; inside a tie the simpler setting won.

Candidate	V loss	Change vs current best	Decision
C0 registered linear (log1p target, numeric weekday, no penalty)	1.28856	—	start
C1 weekday as a category	1.29170	−0.24%	rejected
C2 raw-days target	1.28932	−0.06%	rejected
C3 L1 penalty, best of 3 alphas (0.0001)	1.28713	+0.11%	rejected (< 1%)
D3 HGB, depth 2, 400 iterations (best of 8)	1.26266	—	selected; tie set D3, D4, D7

Post-hoc intervals on V. A week-block bootstrap on V (14 weeks, 2,000 draws) gives an improvement over C0 of 0.11% (−0.002% to 0.245%) for the C3 pick and 2.01% (1.11% to 2.85%) for D3. No improvement in the first interval reaches the pre-set 1% bar, so the original choice stands. Both candidates were picked on this same V, and the intervals leave out selection and refitting uncertainty, so D3's figure is optimistic and was only a reason to evaluate it.

S2. A small simulation of two calibration failures (synthetic data)

Question: does a calibration shift fail because the delivery distribution changes over time, or because at the freeze only the fast, already-delivered orders can be used? Delivery days follow $\log Y = 2.2 + 0.5x + 0.45\varepsilon$, rounded up. The predictor is the conditional median,

and the calibration code is the project's. Each of 1,000 replications draws 2,000 calibration and 5,000 evaluation orders (seed 20261021). All scenarios share the evaluation sample. For the time-change scenarios, the continuous delivery time is rescaled before rounding.

Scenario minus the stable, complete-label scenario	Coverage change, pts (mean ± MC s.e.)	Middle 90% of replications	Promised days change
Only delivered-by-freeze labels (90-day window)	−1.68 ± 0.015	−2.48 to −0.94	−0.99
Random subset of the same size (diagnostic)	+0.02 ± 0.009	−0.42 to +0.46	+0.02
Later deliveries 15% slower	−5.77 ± 0.014	−6.50 to −5.08	0
Later deliveries 15% faster	+4.26 ± 0.012	+3.66 to +4.90	0

The two mechanisms leave different traces. Selective labels shrink the shift itself, so promised days and coverage fall together; a random subset of the same size does not. A later change in delivery times leaves the promise unchanged and moves only coverage. The "middle 90%" column is the spread of replications, not a confidence interval for the mean. A first, unpaired run (seed 20261020) gave the same picture. This explains mechanisms; it does not show which one, if either, acted in the Olist data.

S3. Post-hoc methodological review

Everything in S3 was computed after the scoring run, in response to the third and fourth reviews. It reuses the frozen per-order predictions on C and E, fits nothing, chooses no policy and changes no registered verdict. Sources: scripts/e1_posthoc_review3.py → outputs/e1/posthoc_r3/ and scripts/e1_posthoc_methods.py → outputs/e1/posthoc_m4/ (checksums in each MANIFEST.sha256). S3b replaces an earlier version of the same analysis that used a "subtract one day" rule, and explains why that rule is not valid in general.

S3a. Diagnostic comparison at matched coverage on E. Each model's shift is moved over E (steps of 0.01 day; whole days for the state rule), and the table reads off the fewest mean days among the steps that reach the target. Whole-day promises make coverage move in steps, so the table shows the coverage actually reached; nothing is interpolated. These curves use E's outcomes; they are not deployable policies.

Target coverage	Linear C*: reached, days	HGB D*: reached, days	State rule: reached, days
90.00%	90.02%, 17.20	90.02%, 16.60	90.75%, 17.99
93.95% (B0)	93.95%, 19.83	93.96%, 19.12	94.45%, 20.99
96.18% (C* frozen)	96.18%, 22.44	96.18%, 21.71	96.49%, 23.99
97.10% (D* frozen)	97.11%, 24.57	97.10%, 23.66	97.33%, 25.99

In this sample and for this family of shift policies, the HGB curve lies 0.60–0.91 days below the linear curve at the four targets in the table. That is not a general ranking of the models. On C at the registered shifts: D* 91.14% with 24.54 days, C* 91.19% with 24.68, B1c 90.30% with 24.49.

S3b. From a continuous bound to a whole-day promise. Let $R_i = Y_i - \hat{f}(X_i)$ on the calibration set, $k = \lceil (n+1)q \rceil$, $s = R_{(k)}$ and $U(x) = \hat{f}(x) + s$. If calibration and evaluation scores are exchangeable, the rank argument gives $P(Y \leq U(X)) \geq q$. A change over time is not covered by this guarantee (Barber et al.). Three ways to turn U into whole days:

- **Round up (registered):** $d = \max\{1, \lceil U \rceil\}$. $Y \leq U$ implies $Y \leq d$. Conservative: it adds slack whenever U is not a whole number.
- **Round down:** $d = \max\{1, \lfloor U \rfloor\}$. Because Y is a whole number, $Y \leq U$ holds exactly when $Y \leq \lfloor U \rfloor$, so the untruncated $\lfloor U \rfloor$ has exactly the coverage event of U. The one-day minimum can only enlarge that event: with $Y = 1$ for every order and forecasts 0, 1, …, 99 ($q = 0.9$), U covers 91% and $\max\{1, \lfloor U \rfloor\}$ covers 100%.
- **Subtract one day:** $d = \max\{1, \lceil U - 1 \rceil\}$. This equals $\lfloor U \rfloor$ when U is not a whole number, and $U - 1$ when it is (before the one-day minimum). Counterexample: every delivery takes 10 days and the forecast is constant at 0.5, so $s = 9.5$ and $U = 10$ for every order. Rounding up or down promises 10 days and covers every order; subtracting one day promises 9 and covers none. The rule is not coverage-preserving in general, but it does not always under-cover either. Without the minimum its event is $Y < U$, which on the calibration set means $R_i < R_{(k)}$: fewer than k orders. The minimum can restore coverage for one-day outcomes: with 100 orders of 1 day and a forecast of 0.5, $U = 1$ and $\max\{1, \lceil 0 \rceil\} = 1$ covers all of them.

Numerical detail: $U = \hat{f} + s$ is a floating-point sum, so a bound that should be a whole number can come out just below it. The code compares against $U \pm 10^{-9}$ in the direction that keeps the event ($\lfloor U + 10^{-9} \rfloor$), as the registered rule already does ($\lceil U - 10^{-9} \rceil$). Tests:

tests/test_posthoc_rounding.py (45 cases: the counterexamples, the event identity on 60 random calibration sets, the one-day minimum, boundary and tolerance cases, and refusal of non-finite or malformed inputs).

Policy	Shift	Rule	Coverage on C	Days on E	Coverage on E
B1c state rule	6.00	round up = round down	90.30%	23.99	96.49%
B1c state rule	6.00	subtract one day	89.08%	22.99	95.92%
B2 earlier regression	6.02	round up	91.17%	22.87	96.45%
B2 earlier regression	6.02	round down	90.01%	21.87	95.87%
C* linear	6.15	round up	91.19%	22.45	96.18%
C* linear	6.15	round down	90.01%	21.45	95.51%
D* HGB	6.46	round up	91.14%	23.67	97.10%
D* HGB	6.46	round down	90.01%	22.67	96.66%

q = 0.90, k = 15,352 of n = 17,056 (k / n = 90.01%). No bound on C or E is below 12.5 days for any of these policies, so the one-day minimum never binds here and the table is unaffected by it. Rounding down meets the rank bound on C for every policy. For the continuous policies, subtracting one day gives the same E numbers as rounding down to two decimals (only one E order, for C*, has a bound within 10^{-6} of a whole number) but reaches only 90.00% on C, below k / n. For the integer-valued state rule it falls to 89.08%.

Two different quantities: the registered promise exceeds U by 0.50 day on average (C* 0.497, D* 0.503), which is the arithmetic part of what was done. Rounding down instead shortens every continuous promise by exactly 1 day, because $\lfloor U \rfloor = \lceil U \rceil - 1$ when U is not a whole number; that is a counterfactual. With every policy rounded down: D* − B1c = −1.32 days at +0.18 points; C* − B1c = −2.54 days at −0.97 points; against B0, D* +2.37 days (+2.71 points) and C* +1.15 days (+1.56 points). These are point estimates; the rule was not registered, and the registered H3 verdict stands.

Correction. The third-review version of this table applied "subtract one day" to continuous predictors and left integer predictors unchanged. On these data its E numbers equal those of rounding down, but the rule is not coverage-preserving in general (counterexample above), and on C it reached 90.00% < k / n. It is replaced by rounding down.

S3c. The evaluation sample and its window. The cohort rule keeps orders whose Olist estimated delivery week is no later than 23 July 2018.

Purchase month	Purchases	Estimated week ⩽ 23 Jul 2018	Delivered (on time or late)	Single seller = E
April 2018	6,939	6,939	6,798	6,670
May 2018	6,873	6,862	6,738	6,630
June 2018	6,167	5,732	5,665	5,565
July 2018	6,292	1,117	1,095	1,085
August–October 2018	6,532	0	0	0
Total	32,803	20,650	20,296	19,950

Purchase month (E)	Orders	Mean actual days	Olist promise, mean	Olist promise, max	Late vs Olist
April 2018	6,670	11.47	24.27	63	4.59%
May 2018	6,630	11.36	23.41	60	6.67%
June 2018	5,565	8.92	27.45	56	1.26%
July 2018	1,085	6.22	16.33	26	2.95%

Policy	Full E: coverage, days	April–May: coverage, days	June–July: coverage, days
B0 retained rule	93.95%, 20.31	92.23%, 20.55	97.38%, 19.82
Olist promise	95.74%, 24.44	94.38%, 23.84	98.47%, 25.64
B1c state rule	96.49%, 23.99	95.53%, 24.15	98.39%, 23.67
C* linear	96.18%, 22.45	95.13%, 22.75	98.27%, 21.85
D* HGB	97.10%, 23.67	96.39%, 24.01	98.51%, 22.99

April–May inherits the same cohort rule, so it is a sensitivity check, not a correction. Removing the dependence on Olist's estimated date would need a new purchase-time cohort with its own observation cut-off and handling of unconfirmed deliveries; that is a new design and was not done here. Every calibrated policy still covers well above 90% in April–May.

S3d. Calibration period. Late rates against Olist's date on C, by purchase month: October 2017 4.24% (4,414 orders), November 12.49% (7,191; 42% of C), December 7.43% (5,451). The C shift was therefore learned largely in a peak period. This is a candidate explanation for over-coverage on E; a high November late rate does not by itself show that it caused it, and it was not tested separately.

S3e. Leave one purchase week out. Each of the 17 purchase weeks of E is removed in turn and the registered differences are recomputed from the frozen promises. This is a stability diagnostic, not a confidence interval.

Statistic	Full E	Range with one week removed	Sign kept
H1 gap at R3 (C* − B1c), days	−1.54	−1.60 to −1.49	yes
H3 D* − B1c, days	−0.32	−0.36 to −0.26	yes
H3 D* − B1c, coverage points	+0.61	+0.52 to +0.67	yes
C* − B0, days	+2.15	+2.09 to +2.21	yes
D* − B0, days	+3.37	+3.34 to +3.40	yes
C* (q = 0.9126) − Olist, days	−1.02	−1.41 to −0.35	yes
C* (q = 0.9126) − Olist, coverage points	+0.88	+0.38 to +1.05	yes

The comparison with Olist's promise is the least stable: removing a single week moves its day difference between −1.41 and −0.35.

A. Reproducibility and checks

Item	Value
Pre-registered configuration	config/date_model_e1.yaml v3, SHA-256 ae6a4a46…20152
Freeze record	FREEZE.json, SHA-256 3b62af1a…328dd; 9 runs. The final E1 scoring run came after it (E had been examined in Stage 7)
Code at scoring time	combined fingerprint 6eac633d…af40; a full copy is kept in snapshots/src_at_e1_scoring

Item	Value
Automated tests	345 pass (317 for the registered pipeline, 28 for the post-hoc rounding rules), including synthetic end-to-end tests showing that scoring stops before any E output if the freeze, a model file, the sample, the calibration shifts or (after the post-hoc hardening) any input content changed, or if the post-hoc addendum is malformed or bound to a different freeze. The defect tests were also run against the earlier code to show that they failed there
Independent recomputation	every reported number recomputed from the saved per-order files with plain NumPy / pandas: 43 / 43 for E results, 43 / 43 for supplementary results
Post-hoc hardening	content hashes of all inputs, added after scoring as a labelled addendum bound to the freeze record's hash; the original freeze record is unchanged
Reproduction guide	docs/REPRODUCE_E1.md, recipes A–C; erratum of 2026-10-21 corrects the path used by recipe B (the historical code must sit at src/)
Process record	docs/E1_DECISIONS.md logs every change to the plan with its time, its reason and whether it came from the author or from a review
Environment	Python 3.13.16; numpy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1, scipy 1.18.1
Data	public Olist e-commerce dataset (Kaggle); raw files and order-level marts are not redistributed
Not done	a full rebuild from the raw CSVs by an independent reviewer; a second evaluation period; a cohort that does not depend on Olist's estimated date; a registered study of whole-day rounding rules and of calibration under temporal shift (an E2a protocol is drafted, not run)

B. Contribution statement (draft — the author must check every line)

Part	Who	Author to confirm
Reframing the question from late-order prediction to promise setting	Author, through the project and discussions	[ ]
Experiment scope and constraints (three model families, no tuning on E, staged checkpoints, disclosure rule)	Author	[ ]
Pre-registration text, code for selection, freeze and scoring, tests, simulation, post-hoc checks, the rounding argument and its counterexamples as written here	Drafted and implemented with an AI assistant (Claude), revised over several review rounds	[ ]
Reviews that found design and code issues (comparison confound, freeze gaps, rounding, evaluation window, wording)	Several review rounds by AI reviewers, run at the author's request; not formal peer review [author: name the tools and any human reader]	[ ]
Libraries	scikit-learn (QuantileRegressor, HistGradientBoostingRegressor, mean_pinball_loss definition), NumPy, pandas, SciPy	—
Results I recomputed or re-derived myself	[author: list, e.g. the integer-day derivation, the V check, one E comparison]	[ ]

The lines above should be filled from the actual records, not from memory: docs/E1_DECISIONS.md and the review reports and evidence packages kept with it. The author should be able to explain, without notes: the integer-day derivation; why the calibration shift uses the $(n + 1)q$-th residual; why rounding the bound down keeps its coverage event while subtracting one day does not; why C* and B3 are the same model; and why the frozen HGB and linear policies cannot be ranked from their registered points alone.